Series & Binomial Theorem

Section 1: Arithmetic Progressions (AP)

1 An arithmetic progression has first term $a = 7$ and common difference $d = 4$. Find the 20th term u_{20}.

a. 83

b. 87

c. 79

d. 80

2 Find the sum of the first 25 terms of the AP: $3, 8, 13, 18, \ldots$

a. 1575

b. 1625

c. 1500

d. 1550

3 In an AP, the 5th term is 18 and the 12th term is 39. Find the first term a and common difference d.

a. $a = 6, d = 3$

b. $a = 3, d = 3$

c. $a = 9, d = 3$

d. $a = 6, d = 4$

Section 2: Geometric Progressions (GP)

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4 A geometric progression has first term $a = 5$ and common ratio $r = 2$. Find the 8th term u_8.

a. 640

b. 1280

c. 320

d. 512

5 Find the sum of the first 6 terms of the GP: $3, -6, 12, -24, \ldots$

a. -63

b. 63

c. -127

d. -65

6 The 2nd term of a GP is 12 and the 5th term is 324. Find the common ratio r.

a. 3

b. 2

c. 4

d. $\sqrt{3}$

Section 3: Sum to Infinity of a Convergent GP

7 State the condition for the sum to infinity of a geometric progression to exist.

a. $|r| < 1$

b. $r > 1$

c. $r \leq 1$

d. $|r| > 1$

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8 Find the sum to infinity of the GP: $16, 12, 9, 6.75, \ldots$

a. 64

b. 48

c. 32

d. 72

9 Express the recurring decimal $0.\overline{27} = 0.272727\ldots$ as an exact fraction in lowest terms using an infinite geometric series.

a. $\frac{3}{11}$

b. $\frac{27}{100}$

c. $\frac{9}{33}$

d. $\frac{7}{25}$

Section 4: The Binomial Theorem for Positive Integers

10 Find the coefficient of x^3 in the expansion of $(2 + 3x)^5$.

a. 1080

b. 720

c. 540

d. 1440

11 Find the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

a. 240

b. 60

c. 160

d. 120

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12 In the expansion of $(1 + ax)^6$, the coefficient of x^2 is 60. Given $a > 0$, find a.

a. 2

b. 4

c. 3

d. 5

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